

1(a)(i)	From energy conservation, gain in g.p.e. = loss in k.e. $mg(10.0) = \frac{1}{2}mu^2 - \frac{1}{2}m(5.00)^2$ $u = 14.87 \approx 14.9 \text{ m s}^{-1}.$	M1
(a)(ii)	$14.873 \cos \cos \theta = 5.00$ $\theta = 70.4^\circ$ Or Vertical motion: $u_y = 14.9 \sin \sin q \uparrow$, $a_y = 9.81 \downarrow$, $s_y = 10.0 \uparrow$, $v_y = 0$ From $v^2 = u^2 + 2as$ taking $\uparrow +ve$: $0 = u_y^2 + 2(-9.81)(10.0)$ $u_y = \sqrt{196.2} = 14.873 \sin \sin \theta$ $\theta = 70.4^\circ$	M1 A1 M1 A1
(a)(iii)	Consider vertical motion: $u_y = 14.873 \sin \sin 70.4^\circ \uparrow$, $a_y = 9.81 \downarrow$, $s_y = 0$, $t = ?$ From $s_y = u_y t + \frac{1}{2}a_y t^2$ and taking $\uparrow +ve$: $0 = (14.873 \sin \sin 70.4^\circ)t + \frac{1}{2}(-9.81)t^2$ $0 = \left\{ (14.873 \sin \sin 70.4^\circ) + \frac{1}{2}(-9.81)t \right\} (t)$ $t = 0$ (reject) or $t = \frac{2(14.873 \sin \sin 70.4^\circ)}{9.81} = 2.86 \text{ s}$ or $0 = (14.873 \sin \sin 70.4^\circ) - 9.81t'$ $t = 2t' = 2.86 \text{ s}$	M1 A1
(a)(iv)	Rate of change of momentum = resultant force = weight = mg $= 8.00 \times 10^{-3} \times 9.81$ $= 0.0785 \text{ N}$	A1
(b)	Vertical displacement Correct shape [B1] Time of descent > time of ascent [B1] time	

